NOTRE DAME UNIVERSITY BANGLADESH

Computer Graphics Lab Report-01

Course Code: CSE-4218
Course Title: Computer Graphics Lab
Lab Task Topic: Setting Up Glut & Draw a House

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Submission Date: January 13, 2026

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1 Objective

The objective of this lab is to understand the basics of Computer Graphics programming using OpenGL and GLUT. In this experiment, a simple house structure is created using basic geometric primitives such as quadrilaterals and triangles in a 2D orthographic projection.

2 Tools and Environment

- Programming Language: C++
- Graphics Library: OpenGL with GLUT
- IDE: Code::Blocks
- Operating System: Kali Linux

3 Algorithm

1. Initialize GLUT and create a window.
2. Set the background color and projection using `glClearColor()` and `glOrtho()`.
3. Clear the color buffer.
4. Draw the house base using quadrilaterals.
5. Draw the roof using triangles.
6. Draw door and windows using quadrilaterals.
7. Display the final output on the screen.

4 Code Explanation

The program begins by initializing the OpenGL environment. The `init()` function sets the background color to white and defines a 2D orthographic viewing volume.

The `display()` function is responsible for rendering the house. Different parts of the house such as the base, roof, door, and windows are drawn using OpenGL primitives. Colors are assigned using the `glColor3f()` function.

The main function initializes GLUT, creates a window, registers the display callback, and starts the event loop.

5 Source Code

```
/*
 * Lab Task-01
 * Creating a House using quad line
 * Written by Istiak Alam
 */

#include <GL/glut.h>  // GLUT header is important
#include <stdlib.h>
#include <math.h>
#include <cmath>


// Setup and Projection
void init() {
    glClearColor(1.0f, 1.0f, 1.0f, 1.0f); // White background
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    glOrtho(-10, 10, -10, 10, -1, 1);    // Its just like margine or
        padding
    glMatrixMode(GL_MODELVIEW);
}

void display() {                                        // Display
   function
    glClear(GL_COLOR_BUFFER_BIT);

    glColor3f(1.0f, 0.8f, 0.0f); // Light orange
    glBegin(GL_QUADS);                  // House Base (Rectangle)
        glVertex2f(-5.0f, -3.0f);
        glVertex2f( 5.0f, -3.0f);
        glVertex2f( 5.0f,  3.0f);
        glVertex2f(-5.0f,  3.0f);
    glEnd();

    glColor3f(0.8f, 0.0f, 0.0f); // Dark red
    glBegin(GL_TRIANGLES);              // Roof (Triangle)
        glVertex2f(-6.0f,  3.0f);
        glVertex2f( 6.0f,  3.0f);
        glVertex2f( 0.0f,  7.0f);
    glEnd();

    glColor3f(0.3f, 0.2f, 0.1f); // Brown
    glBegin(GL_QUADS);                  // Door (Rectangle)
        glVertex2f(-1.0f, -3.0f);
        glVertex2f( 1.0f, -3.0f);
        glVertex2f( 1.0f,  0.0f);
        glVertex2f(-1.0f,  0.0f);
    glEnd();
```

```
    glColor3f(0.0f, 0.6f, 1.0f); // Light blue
    glBegin(GL_QUADS);                    // Window 1 (Left)
        glVertex2f(-4.0f,  0.5f);
        glVertex2f(-2.0f,  0.5f);
        glVertex2f(-2.0f,  2.0f);
        glVertex2f(-4.0f,  2.0f);
    glEnd();

    glBegin(GL_QUADS);                    // Window 2 (Right)
        glVertex2f( 2.0f,  0.5f);
        glVertex2f( 4.0f,  0.5f);
        glVertex2f( 4.0f,  2.0f);
        glVertex2f( 2.0f,  2.0f);
    glEnd();

    glFlush();
}

int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutInitWindowSize(800, 900);
    glutCreateWindow("House - Computer Graphics Lab-01");

    init();
    glutDisplayFunc(display);
    glutMainLoop();
    return 0;
}
```

6 Graph Implementation

Figure 1: Graph View

7 Output

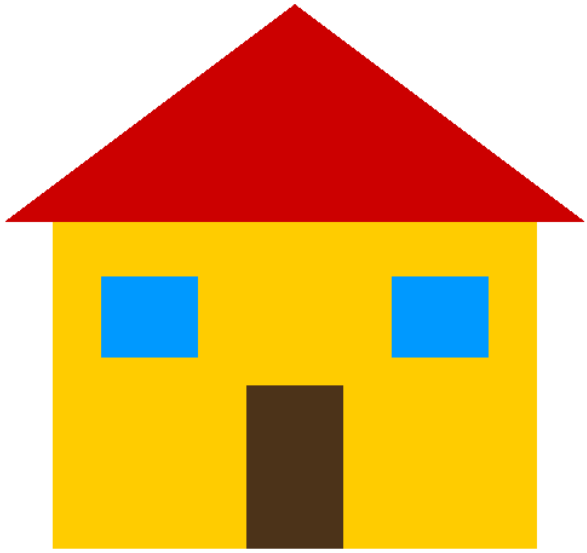

Figure 2: Output of House Drawing Using OpenGL

8 Discussion

This experiment demonstrates how simple geometric shapes can be combined to form complex objects using OpenGL. The use of orthographic projection simplifies the coordinate calculations and is suitable for 2D graphics applications.

9 Conclusion

In this lab, a house was successfully drawn using OpenGL and GLUT. The experiment helped in understanding basic OpenGL functions, coordinate systems, and drawing primitives. This knowledge forms the foundation for more advanced graphics programming.